

SAFETY DATA SHEET

Sulfuric Acid 98%

SECTION 1: IDENTIFICATION

Product Name: Sulfuric Acid 98%

Product Code: SA-98-500

CAS Number: 7664-93-9

Manufacturer: Chemical Supply Corp.

Address: 1234 Industrial Blvd, Houston, TX 77001

Phone: (713) 555-1234

Emergency Phone: CHEMTREC 1-800-424-9300

Recommended Use: Laboratory reagent, industrial processes

Restrictions on Use: Not for household use

SDS Version: 3.2

Revision Date: 2024-01-15

SECTION 2: HAZARD(S) IDENTIFICATION

Signal Word: Danger

GHS Classification: Oxidizing Liquid Category 3, Skin Corrosion Category 1A

H314 Causes severe skin burns and eye damage

H290 May be corrosive to metals

P260 Do not breathe mist, vapors, spray

P264 Wash hands thoroughly after handling

P280 Wear protective gloves/protective clothing/eye protection

Other Hazards: Reacts violently with water (exothermic)

SECTION 3: COMPOSITION / INFORMATION ON INGREDIENTS

Sulfuric Acid 7664-93-9 95-98%

Water 7732-18-5 2-5%

SECTION 4: FIRST-AID MEASURES

Inhalation: Move to fresh air. If breathing is difficult, give oxygen.

Skin Contact: Immediately flush with plenty of water for at least 20 minutes.

Eye Contact: Immediately flush eyes with water for at least 30 minutes.

Ingestion: Do NOT induce vomiting. Rinse mouth with water.

Notes to Physician: Treat symptomatically. Delayed pulmonary edema may occur.

SECTION 5: FIRE-FIGHTING MEASURES

Suitable Extinguishing Media: Use water spray, dry chemical, carbon dioxide, or foam.

Special Hazards: Non-combustible but reacts with metals to produce hydrogen gas.

Protective Equipment: Wear self-contained breathing apparatus.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions: Evacuate area. Wear appropriate protective equipment.

Environmental Precautions: Prevent entry into waterways, sewers, basements.

Containment/Cleanup: Neutralize with soda ash or lime.

SECTION 7: HANDLING AND STORAGE

Handling: Always add acid to water, never water to acid.

Storage: Store in cool, dry, well-ventilated area.

Incompatible Materials: Bases, metals, organic materials, reducing agents, water

SECTION 8: EXPOSURE CONTROLS / PERSONAL PROTECTION

TWA: 1 mg/m³ (OSHA PEL)

STEL: 2 mg/m³ (ACGIH)

Engineering Controls: Use local exhaust ventilation or enclosed process.

Respiratory Protection: NIOSH approved acid gas respirator.

Hand Protection: Chemical-resistant gloves (neoprene, butyl rubber, or PVC).

Eye Protection: Chemical splash goggles and face shield.

Skin Protection: Chemical-resistant apron and boots.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Physical State: Liquid

Color: Colorless to slightly yellow

Odor: Pungent, suffocating

pH: <1 (strongly acidic)

Melting Point: 10.31 C (98%)

Boiling Point: 337 C

Flash Point: Not applicable (non-combustible)

Vapor Pressure: <0.001 mmHg at 20 C

Vapor Density: 3.4 (Air = 1)

Relative Density: 1.84 g/cm³

Solubility: Miscible with water (exothermic)

Molecular Weight: 98.08 g/mol

Molecular Formula: H₂SO₄

SECTION 10: STABILITY AND REACTIVITY

Stability: Stable under normal conditions.

Conditions to Avoid: Heat, moisture, incompatible materials.

Incompatible Materials: Metals, organic materials, reducing agents, bases, water

Hazardous Decomposition Products: Sulfur dioxide, sulfur trioxide fumes.

Possibility of Hazardous Reactions: Violent reaction with water.

SECTION 11: TOXICOLOGICAL INFORMATION

Skin Corrosion/Irritation: Causes severe chemical burns.

Eye Damage/Irritation: Causes severe eye damage.

Carcinogenicity: Strong inorganic acid mists - IARC Group 1

LD50 (oral, rat): 2140 mg/kg

SECTION 12: ECOLOGICAL INFORMATION

Persistence/Degradability: Neutralized in environment by natural buffering.

Bioaccumulative Potential: Not expected to bioaccumulate.

Mobility in Soil: High mobility; dissociates completely in water.

EC50 (Daphnia): 29 mg/L (48h)

SECTION 13: DISPOSAL CONSIDERATIONS

Waste Disposal Method: Neutralize carefully with soda ash or lime before disposal.

EPA Waste Code: D002 (corrosive waste)

Contaminated Packaging: Triple rinse containers and offer for recycling.

SECTION 14: TRANSPORT INFORMATION

UN Number: UN1830

Proper Shipping Name: Sulfuric acid with more than 51 percent acid

Hazard Class: 8

Packing Group: II

DOT Description: UN1830, Sulfuric acid, 8, PG II

SECTION 15: REGULATORY INFORMATION

SARA 311/312: Immediate (Acute) Health Hazard, Reactive Hazard

SARA 313: Listed (Sulfuric Acid, CAS 7664-93-9)

CERCLA RQ: 1000 lbs (454 kg)

RCRA Waste Code: D002

TSCA Status: Listed on TSCA Inventory

California Prop 65: Contains sulfuric acid, known to cause cancer.

SECTION 16: OTHER INFORMATION

Revision Notes: Updated Section 8 exposure limits.

Disclaimer: This SDS is provided without warranty.